

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. MTB-511: Probability

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.

Question 1. fa) Explain the axiomatic definition of: ; ; probability. [2]

(a) Define o-algebra with a suitable . example. — 2 | (ce) Ifdis an empty set, then prove that $P(o)=0$.- 2 (ad) If X is a binomial random variable . mS with $E(X)=2$ and $\text{var}(X)=4$, then evaluate $P(X = 5)$. [2] ' D2(529)—700 (TurnOver)

Question 2. - fe) Define cumulative distribution function. 2 |

(b) A card is drawn at random from a `packofwell-shuffledplayingcards`. Find the probability that it is Spade.

(b) or a King. . 2 | . (g) Define the partition of a sample | | space with an example. 2 : ..(h)
Define $H(X)$ (entropy) and. $M_1(X)$. (sample mean). [2]

(c) Explain $I(X;Y)$ mutual information for the bivariate! - random variables. 2°

(d) Define joint density function and marginal density function for the bivariate random variables. [2]

(e) Let the random variable X has the . . normal 'distribution, — i.e., $Y \sim N(\mu, \sigma^2)$ with parameter μ and σ^2 , then prove that the odd order central moments are zero, i.e., $\mu_r = E[(X - \mu)^r] = 0$, $r = 0, 1, 2, \dots$

$$(f) \quad \begin{array}{cccc} & . & . & . \\ & | & & | \\ & | & & | \end{array}$$

Question 2. (a) Let a random variable X has the - following probability distribution :

(g) Oo 1 2 3 4 5 6 7) oo | “ p(x) O, k 2k 2k 3k k²2k? - 7k? + kThen find (ithe value of k and fi) P(1 - 5 < X < 4 - 5|X > 2) and CDF of X. [4] =
(b) The Cumulative Distribution Function (CDF). of a random variable X is given by oe. 0+ > if x < O x) = 2Ji - gg(8 - 9, if a s x < 3|

(g) Find the Probability Density: ; Function (PDF) of X and evaluate $P(|X| < 1)$ using both PDF and CDF. - 4 ‘

(h) The joint density function of the random variables X and Y is given by $f(x, y) = 6(1-x-y)$, for $x > 0, y > 0, x+y < 1$. Find $F(x, y)$.

Question 8. ; otherwise ' D2(529) — (Turn Over)

- (a) (4) ee then find the marginal density: | functions of X and Y. Verify) whether X and Y are independent | “or not. 4 | |

Question 3. (a) A box contain a white balls and b | *blackballs and a second box, contains c white balls and d black balls. One ball is transferred from |. the first box into the second box' : . and one ball is then drawn from the second box*
 |(b) The probability density function of

random variable X is given by $p(x) = kx(2-x)$, $0 < x < 2$ Then determine the value of k and hence the mean, variance and the [7] distribution function. 4

State and prove the central limit. , theorem. [6]

Question 4. (a) Prove that the Poisson distribution .

(a) a limiting case of the -binomial distribution. [4]

(b) Find the moment generating - function of a random variable. uniformly distributed in the ; interval (a, b) and hence obtain its: . mean and variance. [4]

(c) Let ‘random variable X° with oo exponential distribution i.e., X Exp (4) with parameter i, then find its -mean, variance and moment generating function. . [4]

Question 5. (a) Let X, Y and Z be three jointly

(a) continuous random variables with -

(b) probability density function $k(x+2y+3z)$, $0 < x, y, z < 1$, otherwise 0. ‘Determine the value of k and joint | marginal PDF of X, Y and Z. . - 4 i

(c) If X is the binomial random : : variable with parameter n and p, : then prove that the mean $\mu = np$, | * -, variance $\sigma^2 = npq$ and central moments μ_r , satisfy the recurrence relation : $\mu_r = npq \mu_{r-1} + n p^r - D2(529)(TurnOver)$

Question 6. oy

(d) Prove the memoryless property of | exponential distribution. [4]

Question 6. (a) The joint density function of , random variables X and Y is given — . by . $f(x,y) = Cxy$, $0 < x < 1$, $0 < y < 1$, f(Determine(i) the value of C, |

marginal density function of X and Y, (iii) check whether X and Y are independent of not. and (iv) — ; . $P(X+Y < 1)$. 6ř.

Let the random variable X having a the normal distribution, ie, | $X \sim N(\mu, \sigma^2)$ with parameter μ and σ^2 , then prove that $E(X) = \mu$ and $Var(X) = \sigma^2$. Also determine the ' moment generating function $M_X(t)$. [6]

Question 7. (a) Let the PDF $f(x,y)$ of (X,Y) is | given by | a , $f(x,y) = 8xy$, $0 < x < 1$, $0 < y < 1$

15 f(s y) 0o. Cs otherwise Obtain the marginal PDF of X, the | , conditional PDF of Y given $X = 0.8$: and also evaluate $E(Y|X=0.8)$. [6]

Q. 7) (b) If random variable X has the probability density function $f(x) = te^{-x}$ for $-0 < x < \infty$ Show that the Moment Generating Function (MGF) of X given by $M_X(t) = \frac{1}{1-t}$ —

which exists for $|t| < 1$. Hence, find the mean and variance of X . [6] . tee .